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Reasons for Not Returning to Pre-injury Sport Level After ACL-Reconstruction

Authors
Khaldoon M. Bashaireh[1], Mohammad A. Yabroudi[2], David Logerstedt[3], Lynn Snyder-mackler[4], Zakariya H. Nawasreh[2]

Affiliations
1 Department of Special Surgery, Jordan University of Science and Technology, Irbid central post o, Jordan
2 Division of physical therapy, Dept. of Rehabilitation Sciences, Jordan University of Science and Technology, Irbid, Jordan
3 Physical Therapy, Saint Joseph's University, Philadelphia, United States
4 Physical Therapy, University of Delaware, Newark, United States

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Correspondence
Profs. Khaldoon M. Bashaireh
Jordan University of Science and Technology
Department of Special Surgery
P.O.Box 3030
Irbid central post o
21110 Jordan
Tel.: 00962799051087, Fax: 00962799051087
bashaireh@just.edu.jo

ABSTRACT

The study aimed to identify athlete-reported reasons for not returning to pre-injury sports level after anterior cruciate ligament reconstruction (ACLR) and to identify the factors associated with these reasons. Ninety-one athletes with 2 years post-ACLR indicated whether or not they had returned to their pre-injury sport level (same frequency, duration, and intensity). Athletes who did not return were asked to provide the reasons. Athletes' characteristics and injury-related factors were used to determine factors associated with the reasons for not returning. Only nine athletes (10 %) returned to pre-injury sport level after ACLR. The most common reasons for not returning were lack of confidence or concerns about re-injury (48.8 %), followed by continued post-surgical impairments in the reconstructed knee (39 %). Having episodes of the knee giving way after ACLR was the only significant predictor of post-surgical impairments (48.8 %; OR = 8.3, 95 %CI = 2.48–27.42, p = 0.001). Lack of confidence, concerns about re-injury, or post-surgical impairments in the reconstructed knee were the most frequently reported reasons for not returning to pre-injury sports level with 2 years post-ACLR. Reported dynamic knee instability was the only factor associated with ongoing post-surgical knee impairments after ACLR. Rehabilitation programs should address athletes' psychological responses and resolve knee impairments to optimize return to pre-injury sport level after ACLR.

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Introduction

After rupturing an anterior cruciate ligament (ACL), most athletes opt to undergo ACL reconstruction (ACLR) surgery to restore mechanical knee stability and function, which enables them to return to their pre-injury sport level [1–4]. However, despite the ACLR, the reported rate of return to pre-injury sport level globally is less than optimal and highly variable [5–8], ranging between 33 % and 92 % at approximately 1-year follow-up post-ACLR [6, 7], while the rate of returning to a competitive level of pre-injury sport is as low as 33 % 1-year post-ACLR [5]. A systemic review and meta-analysis of 69 articles demonstrated that 65 % of individuals returned to their pre-injury sport level and only 55 % returned to the pre-injury level of competitive sports at an average of 3.3 years post-ACLR [6]. Thus, it is important to identify the factors that impede the return to pre-injury sport level.

Most studies have used clinical outcomes related to the time from surgery, knee functional performance, psychological status, and patient-reported measures during the process of determining athletes' readiness to return to pre-injury sports levels [6, 9–15]. Furthermore, some studies used these same clinical measures to predict those who could return to their pre-injury sport level at different follow-up times post-ACLR [9, 16–19]. The low rate of return to pre-injury level is associated with older age, poor knee functional performance, limb movement asymmetry, a high psychological response, and a low score on patient-reported measures of knee function [9, 15, 20, 21]. In self-efficacy studies, several athletes are not able to return to their pre-injury sport level despite successful reconstructive surgery and postoperative rehabilitation [22]. The return to pre-injury sport is multifactorial; therefore, it could be that only using clinical outcome measures to assess athletes' readiness and predict returning to pre-injury sport level does not fully explain why many individuals do not return to their pre-injury level. In addition, solely using clinical outcome measures does not account for athletes' perspective on their ability to return to pre-injury sport level [9, 10, 15, 23–27]. Determining potential reasons for not returning to sports might help precisely determining the actual rate of return to sports. Therefore, examining the reasons that account for athletes' perspectives on post-surgical impairments (knee pain, swelling, ROM deficit, instability, and muscle weakness) in the reconstructed knees and their fears of injury recurrence may elucidate the low rate of return to pre-injury sport level. Furthermore, considering factors related to athletes' availability to participate in sports activities or increased work or family obligations after completing the rehabilitation program may influence athletes' decision to return to their pre-injury sport level. As a result, identifying these reasons might help clinicians modify postoperative ACL rehabilitation programs to address these reasons in order to ultimately maximize athletes' rate of return. In addition, identifying the factors related to athletes' availability or increased work or family obligations post-ACLR should be considered when determining athletes' rate of return.

The purpose of this study was to determine the rate of return to pre-injury sport level post-ACLR, to identify the common athlete-reported reasons for not returning to their pre-injury sport level, and to identify the factors associated with these reasons. It was hypothesized that athletes' reasons for not returning would mainly be associated with post-surgical impairments in the reconstructed knee and a lack of confidence with the knee or concerns about injury recurrence.

Materials and Methods

Participants

In May 2017, athletes with primary unilateral hamstring-tendon autograft ACLR were identified from their medical records and contacted to participate in this cross-sectional study between July 2017 and August 2018. Athletes aged 14 to 50 years, who played level I and II sports that included jumping, cutting-pivoting, and lateral movements (i. e. soccer/football, basketball, volleyball, racquetball, gymnastics, and tennis) before their ACL injury and who had primary ACLR 1–5 years earlier, met the inclusion criteria and were asked to participate in the study [28–30].

In addition, athletes who had concomitant meniscus, cartilage, or ligament injuries that did not require surgery and athletes who had meniscus surgery (repair, partial, or complete meniscectomy) were eligible to participate in this study to explore the effect of these concomitant injuries and surgical procedures on their ability to return to their pre-injury sport level. Athletes who had revision ACLR, serious lower extremity injuries (i. e. fractures), prior injuries or surgical procedures to the lower extremity or spine prior to the ACL injury, or other medical conditions or diseases (i. e. cardiopulmonary, or neurological diseases) that affected their ability to return to their pre-injury sport level were excluded. Athletes with concomitant ligamentous or articular cartilage injuries that required surgery were also excluded.

To maximize the generalizability of the results, participation was not restricted to one specific orthopedic surgeon; five senior surgeons who performed the surgeries for the athletes were included in this study. Additionally, the postoperative rehabilitation programs were not standardized for the included athletes. The study protocol was approved by the institutional review board of Jordan University of Science and Technology and King Abdullah University Hospital. All athletes provided an informed written consent form before participating. For participants aged 14–18 years, athletes provided informed written assent, and their parents/guardians provided informed written consent.

Procedures (data collection)

Athletes with ACLR 1–5 years from study participation completed a questionnaire related to their pre-injury sport level participation, knee injury profile (i. e. injured knee, mechanism of injury, concomitant knee injuries), surgical profile (i. e. time from injury to surgery, type of injury), and additional injuries or surgical procedures to the reconstructed or contralateral knees after the primary ACLR. Athletes also indicated whether they have had episodes of the knee giving way since ACLR. All athletes were scheduled to meet with a certified physical therapist to determine their eligibility and to fill out the survey. The questionnaire was self-administered, and athletes answered the questions based on their best understanding.

Athletes indicated their type and frequency of pre-injury sports participation to determine their pre-injury level of activity (see the "Before ACL Injury" section of the appendix). Athletes also answered a global question asking whether they had returned to sports: "*Did you return to sports (yes or no)?*" If yes, "*did you return to your pre-injury sport level, i. e. the same frequency, duration, and intensity of sports (yes or no)?*" (see the "Return to Sports" section of the appendix).

The questionnaire used in this study was first drafted by the coauthors of this study and discussed with therapists, coaching staff, and medical personnel working with sports teams to provide their feedback and corrections. After that, the questionnaire was pilot tested by a small number of injured athletes (n = 10) who had ACLR, and their comments and feedback were integrated in the final draft of the questionnaire. The questionnaire also underwent content and face validity by experts in the field.

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Reasons for not returning to pre-injury sport level

Athletes who did not return to their pre-injury sport level were asked to select one of the following reasons: post-surgical impairments in the reconstructed knee (i. e. pain, swelling, ROM deficit, instability, weakness, etc.); lack of confidence in the knee or worried about re-injury; little time to participate in sports; work or family obligations; and any other reasons (see the "Reasons for Not Returning" section of the appendix). Participants were allowed to choose more than one reason (if applicable) and to rank them according to their importance.

Predictors for the most common reasons for not returning to pre-injury sport level

Pre-injury sport level participation

Athletes who participated in strenuous sports activities involving jumping, cutting, and hard pivoting (i. e. football, basketball, volleyball, or gymnastics) 4–7 times per week were classified as competitive sports players; while those who did not meet these criteria were classified as recreational sports players [31].

Injury profile

Athletes indicated their knee injury side, their dominant side (dominant limb was self-reported as the limb that the player would use to kick a soccer ball for distance), and if they sustained concomitant knee injuries (*yes* or *no*) at the time of the initial ACL injury. The initial ACL injury and concomitant injuries or surgical procedures were confirmed for all participants using magnetic resonance imaging and athletes' surgical reports.

Surgical timing

The time from injury to ACLR was computed in months.

Knee giving-way

Athletes indicated whether they experienced knee giving-way post-ACLR (*yes* or *no*).

Second knee injuries after primary ACLR

Athletes reported whether they sustained a second injury to the same or contralateral knee (*yes* or *no*). This included injury to the graft of the reconstructed knee and ACL of the contralateral knee, meniscus, cartilage, or other knee ligaments individually, or in combination.

Statistical analysis

The rate of return to the pre-injury sport level was computed by dividing the number of athletes who returned to that level by the total number of athletes, with a 95 % confidence interval (CI) determined for the computed rate. An independent t-test was used to determine significant differences in age, body mass index, follow-up time, and time from injury to surgery between those who did and did not return to their pre-injury sport level. The chi-square test was used to determine significant differences in dichotomized variables (pre-injury level of activity, dominant limb, mechanism of injury, episodes of knee giving-way, and second knee injury (same or contralateral knee)) between athletes who did and did not return to their pre-injury sport level. Frequencies and percentages were calculated and summarized for all reasons for not returning to the pre-injury sport level. Frequencies and percentages for reasons that were reported as the only or primary reasons were also quantified.

A univariate logistic regression analysis was used to determine the predictors for the most common reasons accounted for not returning to pre-injury sport level. Predictors included pre-injury activity level, concomitant knee injuries, time from injury to ACLR, episodes of the knee giving way, and whether the athletes sustained a second knee injury post-ACLR [20, 32–35]. An alpha level of < 0.05 was considered significant (odds ratios (OR) and 95 % CI). All statistical analyses were performed using SPSS (version 25.0; IBM Corp, Armonk, NY, USA).

Results

In total, 213 athletes were identified from the medical records and screening; however, 39 athletes (30 %) of those who were eligible for study participation did not respond to the study invitation. Furthermore, there were no differences in baseline demographics between those who did and did not respond ($p \geq 0.07$). Ninety-one athletes (mean age: 26.8 ± 6.6, 95 %CI: 25.4, 28.2) with 2 years (mean follow-up 1.8 ± 1.1 year) post-ACLR participated in this study (▶ **Fig. 1**).

The rate of return to the pre-injury sport level 2 years post-ACLR was 10 % (n = 9, 95 % CI: 4.6–17.9 %). There were no significant differences in the demographic and injury profiles between those athletes who did and did not return to their pre-injury sport level ($p \geq 0.08$). Concomitant injuries, episodes of knee giving-way post-ACLR, and second knee injuries post-ACLR in the ipsilateral and contralateral limbs are summarized in ▶ **Table 1**.

Reasons for not returning to pre-injury sport level

Eighty-two athletes indicated their reasons for not returning to their pre-injury sport level 2 years post-ACLR. Of these, the most reported reason was a lack of confidence or fear of reinjury (n = 56, 68.3 %), with 40 athletes (48.8 %) indicating a lack of confidence or fear of reinjury as the only or primary reason. The second most reported reason was post-surgical impairments in the reconstructed knee (n = 43, 52.4 %), with 32 athletes (39 %) indicating that post-surgical impairments in the reconstructed knee as the only or primary reason for not returning to sports 2 years post-ACLR. Little time to participate in sport (n = 19, 23.2 %) and work or family obligations (n = 19, 23.2 %) was the third most common reason, with 5 athletes (6.1 %) from each response indicating these as the only or primary reason. Two athletes (2.4 %) reported other reasons (one athlete did not complete the rehabilitation, and one had a hamstring muscle injury).

The univariate logistic regression model indicated that there was no significant predictor for lack of confidence in the knee or being concerned about injury recurrence ($p \geq 0.05$); however, concomitant knee injuries was close to being significant (OR:2.71, 95 %CI: 0.98–7.51; p = 0.05) (▶ **Table 2**).

The univariate logistic regression model also indicated that only having episodes of the knee giving way (*yes* or *no*) was a significant predictor of post-surgical impairments in the reconstructed knee

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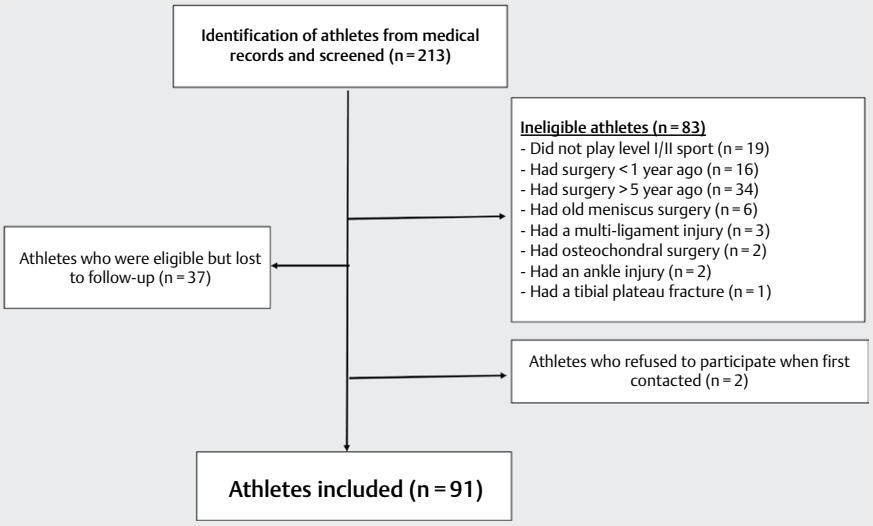

▶ **Fig. 1** Flow diagram of athletes' participation in this study.

as the reason for not returning to pre-injury sport level 2 years post-ACLR (▶ **Table 3**).

Discussion

The current study found that the rate of return to pre-injury sport level 2 years post-ACLR was only 10 %, which is lower than what has been reported globally [6, 9, 15, 36, 37]. Additionally, the findings provided an insight into the commonly reported reasons that contribute to the low return rate. In this study, there was a high percentage of athletes who indicated that they were not psychologically ready to return to pre-injury sport level or who had a fear of injury recurrence, ongoing knee impairment, or functional deficits. In addition, a changed lifestyle and social status influenced almost one-third of the athletes who indicated having less time to participate in sports and increased work or family obligations, which was higher in this sample than reported previously in other studies [38, 39].

Reasons for not returning to pre-injury sport level

The main reasons for not returning to pre-injury sport level were fear of injury recurrence and a lack of confidence in the reconstructed knee. Several studies have shown similar results, with fear of injury recurrence cited as the most common reason for not returning to sports or for returning to a lower sport level post-ACLR [22, 38, 40, 41]. Fear of injury recurrence and lack of confidence could hinder athletes' decision to return to sport post-ACLR; however, it could potentially be modified during postoperative ACL rehabilitation programs. Clinicians should consider this issue post-ACLR to improve athletes' ability to return to sport. Evaluating psychological status during rehabilitation may help identify those with a high fear of injury recurrence and low confidence [42]. The possible reasons that increase athletes' fear and lower their confidence should be discussed, and strategies and techniques should be implemented to minimize fear and improve their confidence throughout the rehabilitation program. This can be achieved by educating athletes about the negative impact of fear on their ability to return to sport and informing them about the expected time to resume functional activities and to return to specific activities; thus, athletes may not feel disappointed and frustrated with their functional performance. In addition, performing regular evaluations of athletes' physical status and functional performance and discussing athletes' progress and improvement might decrease their fears and increase their confidence. Furthermore, having athletes participate in training that progressively and systematically exposes them to activities that place higher demands on the knee may also allow them to develop strategies to perform these activities safely and more confidently, thus reducing their level of fear and improving their chances of returning to sports. Additionally, some athletes may have low personal coping skills to deal with the psychological aspects of the injury [43]. Athletes may also be advised to seek counseling with specialized personnel in sports medicine.

After the fear of injury recurrence and lack of confidence, a common reason for not returning to sport was having ongoing post-surgical impairments in the reconstructed knee. This suggests that it may be worthwhile to consider athletes' opinions on the status of their knee impairment throughout the rehabilitation process. Athletes' knee giving-way post-ACLR was the only significant predictive factor in this category associated with not returning. Possible causes for this might be muscle weakness or a lack of neuromuscular control [10, 15, 23–25, 44–46]. These factors can be addressed during postoperative ACL rehabilitation programs, and clinicians should ensure that knee impairment issues are resolved and should include exercises that help to improve dynamic knee stability. However, other post-surgical complications owing to technical operative issues and post-surgical infection should not be excluded [47].

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▶ **Table 1** Athletes' demographics and injury profiles per returning (or not) to their pre-injury sport level.

Variable	Returned (n = 9)	Did not return (n = 82)	P-value
Age (years), mean ± SD	28.5 ± 7.1	26.6 ± 6.6	0.44
Number of athletes (Men = 89, Women = 2)	(9/0)	(80/2)	0.64
Baseline activity level			
Competitive (n = 38)	6 (66.7 %)	32 (39 %)	0.11
Non-competitive (n = 53)	3 (33.3 %)	50 (60.1 %)	
Follow-up time (years), mean ± SD, years	1.6 ± 0.7	1.8 ± 1.1	0.71
Body mass index, mean ± SD, kg/m^2	24.6 ± 3.9	24.8 ± 3.9	0.86
Injured knee (right)	7 (77.8 %)	39 (47.6 %)	0.09
Injured side (dominant)	6 (66.7 %)	45 (54.9 %)	0.52
Concomitant injury			
Yes (n = 51)	5 (55.6 %)	46 (56.1 %)	0.91
Concomitant injuries (specified)			
Meniscus injury (n = 30)	5	25	
cartilage injury (n = 9)	0	9	
Meniscus and ligament injuries (n = 4)	0	4	
Meniscus and cartilage injuries (n = 4),	0	4	
Other ligament injuries (n = 4)	0	4	
Mechanism of injury			
Contact (n = 35)	6 (66.7 %)	29 (35.8 %)	0.7
Non-contact (n = 55)	3 (33.3 %)	52 (64.2 %)	
Missing (n = 1)			
Time from injury to surgery (months), mean ± SD	3.4 ± 2.2	12.8 ± 15.4	0.09
Second knee injury post-ACLR*	1 (11.1 %)	21 (25.6 %)	0.32
Same knee (n = 8)	0 (0 %)	8 (9.8 %)	
Graft rupture(n = 2)	0	2	
Meniscus (n = 5)	0	5	
Non-specific (n = 1)	0	1	
Contralateral knee (n = 14)	1 (11.1 %)	13 (15.9 %)	0.78
ACL rupture (n = 4)	0	4	
ACL and meniscus (n = 1)	0	1	
Meniscus (n = 3)	0	3	
Meniscus and other ligament (n = 1)	0	1	
Meniscus and baker cyst (n = 1)	0	1	
Cartilage (n = 2)	0	2	
Other ligaments (n = 1)	1	0	
Patellar fracture (n = 1)	0	1	
Episodes of knee giving way post-ACLR (yes = 25)	0 (0 %)	25 (29.3 %)	0.08

SD, standard deviation. *Three athletes sustained second injuries to the same and contralateral knees.

▶ **Table 2** Univariate logistic regression model: Lack of confidence or fear of reinjury as the reason for not returning.

Predictor	Lack of confidence in knee or fear of reinjury (n = 56)	OR	95 % CI	P-value
Baseline activity level (competitive)	20 (35.7 %)	1.54	0.60–3.97	0.37
Concomitant injury (yes)	27 (48.2 %)	2.71	0.98–7.51	0.05
Time from injury to surgery (months), mean ± SD	11.04 ± 12.90	0.98	0.95–1.01	0.17
Knee giving way (yes)	14 (25 %)	0.45	0.16–1.21	0.11
Second knee injury post-surgery (yes)	11 (19.6 %)	0.55	0.19–1.59	0.27

OR, odds ratio; CI, confidence interval; SD, standard deviation.

▶ **Table 3** Univariate logistic regression model: Post-surgical complications with the reconstructed knee as the reason for not returning.

Predictor	Post-surgical complications in the reconstructed knee (n = 43)	OR	95 % CI	P-value
Baseline activity level (competitive)	16 (37.2 %)	1.17	0.48–2.85	0.72
Concomitant injury (yes)	27 (62.8 %)	1.6	0.65–3.91	0.30
Time from injury to surgery (months), mean ± SD	13.4 ± 16.1	1.0	0.98–1.04	0.68
Knee giving way (yes)	**21 (48.8 %)**	**8.3**	**2.48–27.42**	**0.001**
Second knee injury post-surgery (yes)	12 (27.9 %)	1.77	0.62–5.08	0.29

Boldfaced *P* indicates a significant difference. OR, odds ratio; CI, confidence interval; SD, standard deviation.

Predictors for the most common reasons for not returning to pre-injury sport level

Our findings also indicate that episodes of the knee giving-way were the only significant predictor for reported ongoing post-surgical impairments in the reconstructed knee; however, whether the graft ruptured was not verified. This highlights the importance of athletes demonstrating dynamic knee stability before returning to high-demand sports activities. Dynamic knee stability can be addressed during rehabilitation. Post-ACLR, athletes should be trained in a structured rehabilitation program that includes multidirectional mechanical stress and demands on the knee similar to those in sports activities. Improving knee functional stability post-ACLR may increase the likelihood of athletes participating in more sports activity and ultimately maximize their return to pre-injury sport level.

The results of this study revealed that none of the variables, including baseline activity level, concomitant injuries, time from in-

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jury to surgery, knee giving-way, and injury recurrence to the same knee or contralateral knee injury, were associated with fear of injury recurrence or lack of confidence in the reconstructed knee. This may suggest that individuals still fear injury recurrence, even though they do not have post-surgical impairments and regardless of their baseline activity level.

Our study had some limitations. Our cross-sectional study has some inherent recall bias that may have affected the current findings. However, the inclusion criteria ensured all athletes had surgery within 1–5 years to reduce said bias. Second, the sample size was small. Despite several phone calls and reminders to eligible athletes, almost 30 % did not respond or were lost to follow-up, which may have influenced the results. Still, the comparison between those who were enrolled and those who were not yielded non-significant differences in participants' demographics. Third, the findings cannot be generalized to female athletes because only two women athletes participated in this study. The small number of women participants in this study might be attributed to cultural reasons as female participation in sport activities is less common compared to men among our population [48]. Fourth, the mean age of the sample may have influenced the rate of return in this study, as the study sample comprised older individuals. Older athletes tend not to return to their pre-injury sport level participation post-ACLR [38, 39]. Fifth, the rehabilitation program was not standardized among participants. However, it was believed that an unstandardized rehabilitation program might reflect the current clinical practice and maximize the generalizability of the findings. The study included only patients who had primary ACLR with hamstring-tendon autograft, therefore the findings of this study cannot be generalized on patients with other grafts. Furthermore, data related to whether participants received pre-operative rehabilitation and duration of pre-operative rehabilitation program were not collected in this study, which may have an influence on patients' outcomes post-ACLR. Lastly, whether participants' knee giving-way episodes were due to mechanical passive instability was not confirmed by clinical tests or magnetic resonance imaging.

In conclusion, the rate of return to pre-injury sport level 2 years post-ACLR in a sample from a developing country was very low (10 %) compared to that reported globally (65 %). Lack of confidence and fear of re-injury were the most reported reasons for not returning to sports 2 years post-ACLR; followed by post-surgical impairments in the reconstructed knee and then little time to participate in sport, such as work or family obligations. Dynamic knee instability was the only measure that was associated with ongoing impairments in the reconstructed knee as a reported reason for not returning to pre-injury sport level 2 years post-ACLR. During post-operative ACLR rehabilitation programs and when determining whether athletes should return to sport, clinicians should consider athletes' perspectives on knee impairments, psychological status, and time constraints.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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